

Task 31: Binary Tree Level Order Traversal

Problem Description

Given the root of a binary tree, return the level order traversal of its nodes' values. (i.e., from left to right, level by level).

Java Solution

```
class Solution {
    public List<List<Integer>> levelOrder(TreeNode root) {
        List<List<Integer>> res = new ArrayList<>();
        if (root == null) return res;

        Queue<TreeNode> q = new LinkedList<>();
        q.offer(root);

        while (!q.isEmpty()) {
            int size = q.size();
            List<Integer> currentLevel = new ArrayList<>();
            for (int i = 0; i < size; i++) {
                TreeNode curr = q.poll();
                currentLevel.add(curr.val);
                if (curr.left != null) q.offer(curr.left);
                if (curr.right != null) q.offer(curr.right);
            }
            res.add(currentLevel);
        }
        return res;
    }
}
```